

CONTROLLED RESIDUALS IN GAUSSIAN RAMSEY CONSTRUCTIONS

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ABSTRACT. We refine the Gaussian random-graph framework for fixed-ratio Ramsey lower bounds. Relative to the pinned Appendix-B induction of Hunter, Milojević and Sudakov and the exact cumulant input of Lin and Niu, we introduce a nonexchangeable triangular weighted ledger with a deterministic boundary residual. Exact gradient merge uniquely fixes the triangular array and gives only a $(96 \log C)^{-1}$ leading contribution. By retaining the full-box strong-concavity term, we pay a controlled failure of exact merge by square completion and obtain an additional term

$$\hat{H}_*(C) = \frac{1 + o(1)}{64 \log C} > 0.$$

For every sufficiently large fixed C , the resulting source-relative bound is

$$\liminf_{\ell \rightarrow \infty} \frac{1}{\ell} \log R(\ell, \lfloor C\ell \rfloor) \geq -\frac{1}{2} \log p_C + \frac{B_R(C)}{2} + G_*(C) + \hat{H}_*(C).$$

The threshold in C is existential and non-effective. We also identify a narrow scalar residual class for which the coefficient $1/64$ is optimal; the statement does not cover adaptive or higher-order potentials.

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1. INTRODUCTION

The probabilistic argument of Erdős gives the classical exponential scale for diagonal and fixed-ratio Ramsey lower bounds [Erd47]. Ma, Shen and Xie recently obtained the first exponential improvement for $R(\ell, C\ell)$ at every fixed $C > 1$ [MSX25]. Hunter, Milojević and Sudakov gave a Gaussian random-graph construction and stronger quantitative estimates [HMS25]; Lin and Niu sharpened the truncated Gaussian cumulant analysis [LN26]. Truncated Gaussian variance proxies and cumulants also have an independent probabilistic literature [BMA24].

This paper works inside the explicit HMS Appendix-B induction. The new ingredient is not history conditioning by itself—that is already present in the cited constructions—but a deterministic nonexchangeable triangular mean ledger. A boundary derivative must normally match a projection weight exactly. We first show that this exact-merge requirement uniquely determines the array and leaves no scalar optimization freedom. We then retain the negative quadratic term from strong concavity and intentionally permit a small deterministic mismatch. Square completion pays this residual and creates a larger net gain.

1.1. Main theorem. For $C > 1$, let $p_C \in (0, 1/2)$ be the unique solution of

$$C = \frac{\log p_C}{\log(1 - p_C)}.$$

The source quantities $B_R(C)$ and $G_*(C)$ are defined precisely in Section 2. The first is the red-side source ledger term; the second is the previously reviewed exact-CGF improvement.

Theorem 1.1 (Source-relative fixed-ratio bound). *There exists $C_0 > 1$ such that, for each fixed $C \geq C_0$,*

$$\liminf_{\ell \rightarrow \infty} \frac{1}{\ell} \log R(\ell, \lfloor C\ell \rfloor) \geq -\frac{1}{2} \log p_C + \frac{B_R(C)}{2} + G_*(C) + \hat{H}_*(C), \quad (1.1)$$

where $\hat{H}_*(C) > 0$ and

$$\hat{H}_*(C) = \frac{1 + o(1)}{64 \log C} \quad (C \rightarrow \infty). \quad (1.2)$$

The public coarsening $B_R(C)/2 \geq 1/20$ may be substituted in (1.1).

The quantifiers are important: C is fixed before $\ell \rightarrow \infty$, and the proof does not give a numerical C_0 . The source contains unspecified $O(\cdot)$ and $o(\cdot)$ constants, so an effective threshold cannot be extracted from the pinned text alone.

1.2. Contributions.

- (i) We prove rigidity of the exact-gradient triangular recursion.
- (ii) We construct an adapted deterministic residual and prove a uniform full-box Hessian bound.
- (iii) We compute the exact multiplicity of the residual and its square penalty.
- (iv) We insert a three-factor Hölder decomposition without counting the old exact-CGF gain twice.
- (v) We propagate the ledger through reverse induction and quantify the perfect-sequence extraction loss.
- (vi) We optimize a narrowly specified scalar residual class and obtain the leading coefficient $1/64$.

The result is stated as a source-relative strengthening. A direct claim of improvement over every printed final normalization in Lin–Niu would require both expressions to be derived from one common terminal probability formula; that comparison is not asserted here.

2. GAUSSIAN SOURCE LEDGER AND NOTATION

We use the frozen notation of the HMS/Lin–Niu bridge. Let $c = c(C)$ be the central cutoff, let $d = d(\ell, C)$ be the Gaussian dimension, and write

$$m(x) = \frac{\varphi(x)}{\Phi(-x)}, \quad m_0 = m(c), \quad m'_0 = m'(c). \quad (2.1)$$

The scale parameter $D = D(C)$ is chosen so that $d \sim D^2 \ell^2$, and we put

$$\rho = \frac{m_0 m'_0}{D}, \quad \omega_0 = \frac{100 \log(10/pC)}{D}. \quad (2.2)$$

For fixed $w > \omega_0$, define

$$M_w = m(c + w), \quad j_-(w) = m'(c - w), \quad j_+(w) = m'(c + w), \quad (2.3)$$

and

$$\theta = m_0^2, \quad A_0 = m_0^2 m'_0. \quad (2.4)$$

Set

$$a_- = \min\{A_0, \theta\}, \quad a_+ = \max\left\{\theta, \frac{A_0 + \rho\theta}{1 - \rho}\right\}, \quad (2.5)$$

$$\mu_w = j_-(w) - j_+(w)^2 \frac{m_0 + \theta/D}{D(1 - \rho)}, \quad (2.6)$$

and

$$\hat{L}_w(C) = \frac{4m_0 M_w^2}{D} + \frac{2M_w^2 a_+}{D^2} + \frac{32m_0^2 M_w^2 a_+}{a_- D^2}. \quad (2.7)$$

Finally,

$$\hat{B}_w(C) = m_0^2 - \frac{16}{3} - \hat{L}_w(C), \quad E_w(C) = \hat{B}_w(C)\theta - \frac{\theta^2}{2\mu_w}, \quad (2.8)$$

$$\hat{H}(C, w) = \frac{\hat{B}_w(C)A_0 + E_w(C)}{12D^2}, \quad \hat{H}_*(C) = \lim_{w \downarrow \omega_0} \hat{H}(C, w). \quad (2.9)$$

Source interface 2.1 (Pinned source interface). We use the HMS v2 Gaussian construction, its Appendix-B reverse induction, the red perfect-sequence extraction, and the blue comparison at the exact locations identified in the reproducibility manifest. We also use the reviewed exact one-sided truncated-Gaussian CGF deficit $G_*(C)$, inserted before the HMS coarsening. These inputs supply a frozen envelope $\mathcal{H}_C(r)$, a source error constant K , and the unchanged first-moment conversion from red/blue probability bounds to (1.1).

No source $O(\cdot)$ term is assigned a numerical constant in this paper. The construction below only deletes additional mass from the source envelope and therefore uses the same existential K .

3. RIGIDITY OF EXACT GRADIENT MERGE

For a target ordered set $[r]$, consider deterministic triangular weights W_{ij} and the connection-plus-mean potential

$$F_i(b) = \sum_{j>i} \log \Phi(-b_j) - \frac{1}{d} \sum_{i<j<q} W_{jq} m(b_j) m(b_q). \quad (3.1)$$

At the central cutoff,

$$\partial_j F_i(c\mathbf{1}) = -m_0 - \frac{m_0 m'_0}{d} \sum_{\substack{q>i \\ q \neq j}} W_{jq}. \quad (3.2)$$

Proposition 3.1 (Unique exact-merge array). *If the linear term in (3.2), multiplied by $b_j - c = \sqrt{d} h_{ij}$, must equal $-W_{ij} h_{ij}$ at every boundary, then the weights are uniquely determined by*

$$W_{ij} = m_0 \sqrt{d} + \frac{m_0 m'_0}{\sqrt{d}} \sum_{\substack{q>i \\ q \neq j}} W_{jq}. \quad (3.3)$$

Uniform or edge-dependent potential rescaling creates no additional effective degree of freedom.

Proof. The coefficient comparison gives (3.3). Process the first endpoint in decreasing order. Every weight on the right side has already been defined, so the system is triangular and has a unique solution. If the potential is scaled by κ , set $W = \kappa T$; the same coefficient comparison again yields (3.3) for W . Edge-dependent scalar factors likewise only rename the effective coefficients. $\square$

The feedback operator in (3.3) has row norm at most $\rho = m_0 m'_0 / D = o(1)$. Its accumulated increment is

$$2A_0 \binom{r}{4} [1 + O(\rho)]. \quad (3.4)$$

With the source centered coefficient $m_0^2[1 + o(1)]$, the exact-merge ledger contributes the previously reviewed term $(1 + o(1))/(96 \log C)$. Exact feedback evaluation can change only the next order.

4. CONTROLLED RESIDUAL AND STRONG CONCAVITY

Set

$$t_0 = m_0 \sqrt{d}, \quad \gamma_d = \frac{m_0 m'_0}{\sqrt{d}}, \quad e_{ij} = \theta(i - 1). \quad (4.1)$$

In decreasing order of the first endpoint define

$$T_{ij} = t_0 + \gamma_d \sum_{\substack{q > i \\ q \neq j}} T_{jq} + e_{ij}. \quad (4.2)$$

The array is deterministic and adapted: when column i is exposed, every future weight in (4.2) has already been fixed.

With F_i as in (3.1), now using T , we obtain the exact residual identity

$$\partial_j F_i(c\mathbf{1}) = -\frac{T_{ij} - e_{ij}}{\sqrt{d}}. \quad (4.3)$$

Thus the mismatch is explicit and deterministic.

Lemma 4.1 (Uniform Hessian bound). *For all sufficiently large fixed C , every $b \in [c - w, c + w]^{V_i}$ satisfies*

$$-\nabla^2 F_i(b) \succeq \mu_w I, \quad (4.4)$$

where $\mu_w > 0$ is defined in (2.6).

Proof. The recursion gives

$$T_{\max} \leq \frac{t_0 + \theta r}{1 - \rho}, \quad \frac{1}{d} \max_j \sum_{q \neq j} T_{jq} \leq \frac{m_0 + \theta/D}{D(1 - \rho)}.$$

The diagonal of $-\nabla^2 F_i$ is at least $j_-(w)$. The absolute off-diagonal row sum is at most the second term subtracted in (2.6), multiplied by $j_+(w)^2$. Gershgorin's theorem proves (4.4). $\square$

Write

$$\delta_j = b_j - c = \sqrt{d} h_{ij} + \varepsilon_{ij}, \quad |\varepsilon_{ij}| \leq d^{-1/5}.$$

Strong concavity, (4.3), and

$$\frac{e_{ij} \delta_j}{\sqrt{d}} - \frac{\mu_w \delta_j^2}{2} \leq \frac{e_{ij}^2}{2\mu_w d} \quad (4.5)$$

give

$$F_i(b) \leq F_i(c\mathbf{1}) - \sum_{j>i} T_{ij} h_{ij} + \sum_{j>i} \frac{e_{ij}^2}{2\mu_w d} + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j>i} T_{ij}. \quad (4.6)$$

The full boundary coefficient merges into the projection potential, while the failure of exact gradient merge is paid by the square term.

5. DETERMINISTIC MASS AND THE CENTERED-MOMENT LEDGER

Write $\Delta_{ij} = T_{ij} - t_0$. The recursion implies

$$A_0(r-j-1) + \theta(j-1) \leq \Delta_{jq} \leq \frac{A_0 + \rho\theta}{1-\rho}(r-j-1) + \theta(j-1). \quad (5.1)$$

Consequently

$$a_-(r-2) \leq \Delta_{jq} \leq a_+(r-2). \quad (5.2)$$

For a future set V_i of size k , put

$$S_i = \sum_{\{j,q\} \subset V_i} \Delta_{jq}, \quad R_i = \max_{j \in V_i} \sum_{q \neq j} \Delta_{jq}.$$

Then

$$S_i \geq a_-(r-2) \binom{k}{2}, \quad R_i \leq a_+(r-2)(k-1), \quad \frac{R_i}{S_i} \leq \frac{2a_+}{a_-k}. \quad (5.3)$$

Lemma 5.1 (Exact accumulated multiplicities). *Let*

$$K_r = \sum_{i=1}^{r-1} (r-i)(i-1)^2 = \binom{r}{3} + 2\binom{r}{4}. \quad (5.4)$$

Then

$$\sum_{i=1}^{r-1} S_i \geq 2A_0 \binom{r}{4} + \theta K_r, \quad \sum_{1 \leq i < j \leq r} e_{ij}^2 = \theta^2 K_r. \quad (5.5)$$

Proof. Sum the two terms of the lower bound in (5.1) with their exposure multiplicities. The future-degree term counts every ordered quadruple twice, giving $2\binom{r}{4}$; the residual term is exactly the sum in (5.4). The second identity follows directly from $e_{ij} = \theta(i-1)$. $\square$

For the centered comparison retain the predecessor decomposition $T = t_0 + \Delta$ and choose conjugate exponents

$$Q_0 = \frac{d}{4t_0k}, \quad Q_1 = \frac{d}{4R_i}, \quad P^{-1} = 1 - Q_0^{-1} - Q_1^{-1}. \quad (5.6)$$

They are legal for all sufficiently large C because

$$\frac{4m_0}{D} + \frac{4a_+}{D^2} < \frac{1}{2}. \quad (5.7)$$

The weighted quadratic factor costs at most

$$\frac{16}{3} \frac{S_i}{d}. \quad (5.8)$$

The three linear-CGF comparison errors sum to at most $\hat{L}_w(C)S_i/d$. Subtracting these costs from the deterministic central gain $m_0^2 S_i/d$ leaves

$$-\hat{B}_w(C)\frac{S_i}{d}. \quad (5.9)$$

The old deficit $G_*(C)$ is used exactly once. It compares the uniform t_0 centered-linear term with the pre-Cauchy source allocation. Equation (5.9) pays only the new Δ factor and the change to the three-factor Hölder exponent, while (4.6) pays the residual. These ledgers are disjoint.

6. REVERSE INDUCTION AND PERFECT-SEQUENCE EXTRACTION

At exposure step i , append to the frozen one-step exponent

$$-\mathbf{d}_{k,w,d}(P_{k,d}^*) - \hat{B}_w(C)\frac{S_i}{d} + \sum_{j>i} \frac{e_{ij}^2}{2\mu_w d} + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j>i} T_{ij}. \quad (6.1)$$

Every coefficient is fixed before the relevant column is sampled. The projection identity retains each future edge, and (4.6) supplies every boundary edge with its full coefficient.

At the empty state, Lemma 5.1 yields

$$P_{R,r}^* \leq \mathcal{H}_C(r) \exp \left[-\mathcal{D}_{C,w,d}(r) - \frac{1}{d} \left\{ 2\hat{B}_w A_0 \binom{r}{4} + E_w K_r \right\} + J_{r,d} \right], \quad (6.2)$$

where

$$0 \leq J_{r,d} \leq \frac{m_0 + \theta/D}{1 - \rho} d^{-1/5} \binom{r}{2} = o_C(r^2). \quad (6.3)$$

For perfect-sequence extraction, relabel every retained ordered set of size q and rebuild the deterministic recursion with target size q . All feasibility inequalities improve because $q/\sqrt{d} \leq 1/D$. If u vertices are deleted, restoring the new deficit at size ℓ costs at most

$$\frac{1}{d} \left[2\hat{B}_w A_0 u \binom{\ell}{3} + E_w \left(u \binom{\ell}{2} + 2u \binom{\ell}{3} \right) \right] \leq \left[\frac{\hat{B}_w A_0 + E_w}{3D^2} + o_C(1) \right] \ell u. \quad (6.4)$$

The bracket is $O(1/\log C) < 1$. The unchanged source failure factor $(p/10)^{10\ell u}$ dominates (6.4), the old extraction costs and (6.3). Summing over deleted subsets remains $o(1)$. Thus the same source first-moment argument applies with the new exponent intact.

7. RATE CONVERSION AND ASYMPTOTICS

Let $C \rightarrow \infty$ after the fixed- C theorem and then let $w \downarrow \omega_0$. The source asymptotics give

$$\rho = o(1), \quad m_0' \rightarrow 1, \quad M_w/m_0 \rightarrow 1, \quad j_{\pm}(w) \rightarrow 1, \quad \mu_w \rightarrow 1, \quad (7.1)$$

$$\hat{L}_w = o(m_0^2), \quad \hat{B}_w = m_0^2[1 + o(1)]. \quad (7.2)$$

Hence all feasibility and positivity conditions used above hold for sufficiently large fixed C . Since $K_\ell/\ell^4 \rightarrow 1/12$, the new contribution to the first-moment rate is exactly (2.9).

Moreover,

$$m_0^2 = (2 + o(1)) \log C, qquad D^2 = (32 + o(1))(\log C)^3, \quad (7.3)$$

and therefore

$$\hat{B}_w A_0 = (4 + o(1))(\log C)^2, qquad E_w = (2 + o(1))(\log C)^2. \quad (7.4)$$

Substitution into (2.9) gives

$$\hat{H}_*(C) = \frac{(6 + o(1))(\log C)^2}{12(32 + o(1))(\log C)^3} = \frac{1 + o(1)}{64 \log C}. \quad (7.5)$$

The exact-merge predecessor is $H_*(C) = (1 + o(1))/(96 \log C)$, so the new term is eventually strictly larger. Because $G_* + \hat{H}_* = O(1/\log C) = o(1)$, the frozen red–blue crossing is unchanged. This proves Theorem 1.1.

7.1. A narrow scalar method cap. Consider residual arrays that satisfy all of the following: they use the same deterministic triangular weighted-pair recursion; their increments are nonnegative and fixed before the relevant column is sampled; they are paid by the same scalar full-box strong-concavity inequality; and they use the same three-factor Hölder/CGF comparison. For such a residual, the direct leading contribution at edge ij is

$$\hat{B}_w(i-1)e_{ij} - \frac{e_{ij}^2}{2\mu_w}. \quad (7.6)$$

Pointwise optimization gives $e_{ij} = \mu_w \hat{B}_w(i-1) = m_0^2(i-1)[1+o(1)]$. Feedback has norm $\rho = o(1)$, so it changes only lower-order terms. Thus $1/64$ is the leading cap for this explicitly admissible scalar class.

This is not a cap on adaptive, matrix-valued or higher-order states, on a different source quadratic MGF, or on a different extraction invariant.

8. REPRODUCIBILITY AND CLAIM BOUNDARY

The proof specification, arithmetic checker and independent source-level review have SHA-256 prefixes **2234a769**, **616040b9** and **ef6ddbce**, respectively. The checker replays the recursion, multiplicity identities, Hessian samples, exact-CGF increments, finite margins at four large C -values and convergence to the leading constant. It is an arithmetic witness, not a substitute for the reverse-induction proof.

From the repository root run

```
bash scripts/bootstrap.sh
```

```
bash scripts/reproduce_lower.sh
```

to reproduce the bridge, exact-CGF predecessor, nonexchangeable ledger and controlled-residual calculations with pinned dependencies.

The theorem is source-relative, fixed- C , asymptotic in ℓ , and non-effective in C_0 . It supplies no new blue-side construction and no finite Ramsey number. The current manuscript does not claim a common-normalization strict improvement over the final printed Lin–Niu theorem, global optimality, priority, publication, or world-best status.

APPENDIX A. DETAILED ONE-STEP LEDGER ESTIMATES

This appendix records the estimates used in Sections 4 and 5 in a form independent of the exploratory calculation log. Throughout, C is fixed and sufficiently large, $w > \omega_0$, and the target set exposed after column i is $V_i = \{i + 1, \dots, r\}$, of size $k = r - i$.

A.1. The Hessian on the entire cutoff box. Recall

$$F_i(b) = \sum_{j \in V_i} \log \Phi(-b_j) - \frac{1}{d} \sum_{\{j,q\} \subset V_i} T_{jq} m(b_j) m(b_q).$$

Since $(\log \Phi(-x))' = -m(x)$, the entries of the Hessian are

$$\partial_{jj}^2 F_i(b) = -m'(b_j) - \frac{m''(b_j)}{d} \sum_{q \in V_i \setminus \{j\}} T_{jq} m(b_q), \quad (\text{A.1})$$

$$\partial_{jq}^2 F_i(b) = -\frac{T_{jq}}{d} m'(b_j) m'(b_q), \quad j \neq q. \quad (\text{A.2})$$

The inverse Mills ratio satisfies $m'' \geq 0$. Hence the interaction term in (A.1) can only strengthen the diagonal of $-\nabla^2 F_i$. On $[c-w, c+w]^{V_i}$, monotonicity gives

$$m'(b_j) \geq j_-(w), \quad |m'(b_j) m'(b_q)| \leq j_+(w)^2. \quad (\text{A.3})$$

The residual recursion and $r/\sqrt{d} \leq 1/D$ imply

$$T_{\max} \leq \frac{t_0 + \theta r}{1 - \rho}, \quad (\text{A.4})$$

$$\frac{1}{d} \max_j \sum_{q \neq j} T_{jq} \leq \frac{m_0 + \theta/D}{D(1 - \rho)}. \quad (\text{A.5})$$

Combining (A.1)–(A.5) with Gershgorin’s theorem proves

$$-\nabla^2 F_i(b) \succeq \left[j_-(w) - j_+(w)^2 \frac{m_0 + \theta/D}{D(1 - \rho)} \right] I = \mu_w I. \quad (\text{A.6})$$

This is a full-box statement, not an expansion valid only at the central cutoff.

A.2. Boundary merge with a controlled residual. The gradient at $c\mathbf{1}$ is

$$\partial_j F_i(c\mathbf{1}) = -m_0 - \frac{m_0 m'_0}{d} \sum_{q \in V_i \setminus \{j\}} T_{jq}. \quad (\text{A.7})$$

The defining recursion

$$T_{ij} = m_0 \sqrt{d} + \frac{m_0 m'_0}{\sqrt{d}} \sum_{q \in V_i \setminus \{j\}} T_{jq} + e_{ij} \quad (\text{A.8})$$

therefore gives

$$\partial_j F_i(c\mathbf{1}) = -\frac{T_{ij} - e_{ij}}{\sqrt{d}}. \quad (\text{A.9})$$

For $\delta_j = b_j - c = \sqrt{d} h_{ij} + \varepsilon_{ij}$, with $|\varepsilon_{ij}| \leq d^{-1/5}$, the tangent inequality from (A.6) gives

$$\begin{aligned} F_i(b) &\leq F_i(c\mathbf{1}) - \sum_{j \in V_i} T_{ij} h_{ij} + \sum_{j \in V_i} \left(\frac{e_{ij} \delta_j}{\sqrt{d}} - \frac{\mu_w \delta_j^2}{2} \right) + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j \in V_i} T_{ij} \\ &\leq F_i(c\mathbf{1}) - \sum_{j \in V_i} T_{ij} h_{ij} + \sum_{j \in V_i} \frac{e_{ij}^2}{2\mu_w d} + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j \in V_i} T_{ij}. \end{aligned} \quad (\text{A.10})$$

The second line is the scalar square completion $xy - \mu_w y^2/2 \leq x^2/(2\mu_w)$. Thus every boundary edge merges with its full coefficient T_{ij} ; only the explicit residual is paid.

A.3. Mass identities. Let $\Delta_{ij} = T_{ij} - t_0$. Backward induction in the first endpoint gives, for every $j < q$,

$$A_0(r-j-1) + \theta(j-1) \leq \Delta_{jq} \leq \frac{A_0 + \rho\theta}{1-\rho}(r-j-1) + \theta(j-1). \quad (\text{A.11})$$

For S_i and R_i as in Section 5, this implies

$$S_i \geq a_-(r-2) \binom{k}{2}, \quad R_i \leq a_+(r-2)(k-1), \quad \frac{R_i}{S_i} \leq \frac{2a_+}{a_-k}. \quad (\text{A.12})$$

The exact accumulated sums are obtained by exchanging the order of summation. The future-degree part of (A.11) counts each ordered quadruple twice, while the residual part contributes

$$K_r := \sum_{i=1}^{r-1} (r-i)(i-1)^2 = \binom{r}{3} + 2\binom{r}{4}. \quad (\text{A.13})$$

Consequently

$$\sum_{i=1}^{r-1} S_i \geq 2A_0 \binom{r}{4} + \theta K_r, \quad \sum_{1 \leq i < j \leq r} e_{ij}^2 = \theta^2 K_r. \quad (\text{A.14})$$

No asymptotic replacement is used in (A.13)–(A.14).

A.4. Three-factor Hölder comparison. Decompose the centered exponent into the source uniform part, the weighted increment, and the remaining factor. Use conjugate exponents

$$Q_0 = \frac{d}{4t_0k}, \quad Q_1 = \frac{d}{4R_i}, \quad P^{-1} = 1 - Q_0^{-1} - Q_1^{-1}. \quad (\text{A.15})$$

The source square-moment estimate is applicable because

$$\frac{4m_0}{D} + \frac{4a_+}{D^2} < \frac{1}{2}. \quad (\text{A.16})$$

The weighted quadratic factor is bounded by

$$\frac{16}{3} \frac{S_i}{d}. \quad (\text{A.17})$$

The three changes in the linear cumulant comparison are, respectively, bounded by

$$\frac{4m_0M_w^2}{D} \frac{S_i}{d}, \quad (\text{A.18})$$

$$\frac{2M_w^2a_+}{D^2} \frac{S_i}{d}, \quad (\text{A.19})$$

$$\frac{32m_0^2M_w^2a_+}{a_-D^2} \frac{S_i}{d}. \quad (\text{A.20})$$

For (A.20), one first uses the row-sum bound R_i , then $\sum_j (u_j^0)^2 \leq 2m_0^2M_w^2S_i/a_-$; this is precisely where the ratio R_i/S_i in (A.12) enters. Adding (A.18)–(A.20) gives $\widehat{L}_w(C)S_i/d$. The central deterministic gain is $m_0^2S_i/d$, and hence the net increment is

$$-\left(m_0^2 - \frac{16}{3} - \widehat{L}_w(C)\right) \frac{S_i}{d} = -\widehat{B}_w(C) \frac{S_i}{d}. \quad (\text{A.21})$$

The pre-existing exact-CGF term $G_*(C)$ compares the uniform t_0 centered component with the source allocation. Equations (A.17)–(A.21) concern only $\Delta = T - t_0$ and the change in Hölder exponent. Therefore $G_*(C)$ and the new term are disjoint ledger entries.

A.5. One-step conclusion and extraction. Combining (A.10), (A.14), and (A.21) gives the empty-state bound

$$P_{R,r}^* \leq \mathcal{H}_C(r) \exp \left[-\mathcal{D}_{C,w,d}(r) - \frac{1}{d} \left\{ 2\widehat{B}_w A_0 \binom{r}{4} + E_w K_r \right\} + J_{r,d} \right], \quad (\text{A.22})$$

where

$$0 \leq J_{r,d} \leq \frac{m_0 + \theta/D}{1 - \rho} d^{-1/5} \binom{r}{2} = o_C(r^2). \quad (\text{A.23})$$

If extraction deletes u vertices from a retained set of size ℓ , then the loss in the new deterministic exponent is at most

$$\frac{1}{d} \left[2\hat{B}_w A_0 u \binom{\ell}{3} + E_w \left(u \binom{\ell}{2} + 2u \binom{\ell}{3} \right) \right] \leq \left[\frac{\hat{B}_w A_0 + E_w}{3D^2} + o_C(1) \right] \ell u. \quad (\text{A.24})$$

The bracket is $O(1/\log C) < 1$. The unchanged source failure charge is exponential in $-\ell u$ with a larger absolute constant, so the sum over deleted subsets is $o(1)$. This proves that (A.22) survives the same perfect-sequence extraction as the pinned source theorem.

APPENDIX B. PINNED SOURCE BRIDGE AND QUANTIFIER LEDGER

This appendix states exactly what is imported from the Gaussian source construction and what is proved locally. It is included to make the claim boundary auditable without relying on status labels in the research log.

B.1. Version-pinned inputs. The source snapshots are Hunter–Milojević–Sudakov v2 [HMS25] and Lin–Niu v2 [LN26]. Their source hashes, filenames, and retrieval URLs are recorded in the repository manifest. The imported ingredients are:

- (S1) the HMS Gaussian construction and Appendix-B reverse induction;
- (S2) the red perfect-sequence extraction and the original blue bound;
- (S3) the centered square-moment estimate in the range used in (A.16);
- (S4) the one-sided truncated-Gaussian cumulant identity used to define $G_*(C)$; and
- (S5) the first-moment conversion from red and blue probability exponents to a Ramsey lower bound.

The local argument changes none of these statements. It adds a deterministic negative term to the red ledger, checks that extraction retains it, and then uses the same red–blue comparison.

B.2. Literal red-side bridge. For orientation, the preceding source audit establishes a red deficit of the form

$$\mathcal{D}_C(r) = \frac{c_R(C)}{d} \sum_{k=1}^{r-1} k(k-1)^2, \quad c_R(C) > 0, \quad (\text{B.1})$$

inside the original Appendix-B recursion. Since

$$\sum_{k=1}^{r-1} k(k-1)^2 = \frac{r(r-1)(r-2)(3r-5)}{12}, \quad (\text{B.2})$$

the corresponding rate term is $c_R(C)/(4D^2)$. This bridge proves that a same-history, pre-coarsening deficit can be carried through the public induction and extraction. The optimized exact-CGF term $G_*(C)$ used in the present paper is the reviewed refinement of that bridge.

The blue-side construction is not modified here. For sufficiently large fixed C , its source exponent remains strictly above the red exponent, because every added red term is $O(1/\log C) = o(1)$ whereas the frozen red–blue gap is of constant order before the final $C \rightarrow \infty$ limit. Thus the red side remains the bottleneck.

B.3. Order of quantifiers. The source $O(\cdot)$ and $o(\cdot)$ terms do not carry published numerical constants. The formal theorem therefore has the order

$$\exists C_0 \forall C \geq C_0 \exists \ell_0(C) \forall \ell \geq \ell_0(C), \quad (\text{B.3})$$

followed by the limit $\ell \rightarrow \infty$. Only after this fixed- C theorem is established do we take $C \rightarrow \infty$ to obtain the coefficient $1/64$. No explicit value of C_0 or $\ell_0(C)$ is asserted.

The source error constant K is frozen before C_0 is selected. The new ledger deletes probability mass relative to the source envelope and adds only the explicitly bounded deterministic error $J_{r,d}$ in (A.23). Therefore no new, history-dependent source constant is introduced.

B.4. Claim boundary. The theorem is source-relative in two senses. First, it is conditional on the version-pinned interfaces (S1)–(S5). Second, comparison with a printed final constant in another normalization requires translating both estimates from the same probability inequality. We have not used such a translation to make a priority or best-known claim. What is proved here is the internal strengthening

$$G_*(C) + H_*(C) \longrightarrow G_*(C) + \hat{H}_*(C), \quad \hat{H}_*(C) > H_*(C) \quad (\text{B.4})$$

for all sufficiently large fixed C , together with $\hat{H}_*(C) = (1 + o(1))/(64 \log C)$.

The frozen arithmetic checker independently verifies the triangular recursion, the exact K_r identity, the full-Hessian inequalities at representative large values of C , the strict positivity margins, and the asymptotic coefficient. It does not replace the probabilistic induction; that induction is given in Section 6 and Appendix A.

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